

Line, Surface, and Volume Elements

Thursday, July 3, 2025 3:01 AM

24. Line, Surface and Volume Elements

24.1 Recall that for coordinate transformation

$$(x_1, x_2, x_3) \rightarrow (q_1, q_2, q_3)$$

we can calculate Lamé coefficients: h_1, h_2, h_3 , where

$$h_j^2 = \sum_{k=1}^3 \left(\frac{\partial x_k}{\partial q_j} \right)^2.$$

Today, we will use these Lamé coefficients to compute surface areas and volumes in cylindrical and spherical coordinates.

24.2 Today's Hero

Line element:

metric form: $ds^2 = h_1^2 dq_1^2 + h_2^2 dq_2^2 + h_3^2 dq_3^2$

differential form: $ds = h_1 dq_1 + h_2 dq_2 + h_3 dq_3$

Note: Think ds more as a vector and ds^2 as its (length)².

In Cylindrical Coordinates:

$$ds^2 = dr^2 + r^2 d\theta^2 + dz^2$$

$$r^2 = x^2 + y^2$$

In Spherical Coordinates

$$ds^2 = d\rho^2 + \rho^2 \sin^2 \phi d\theta^2 + \rho^2 d\phi^2$$

$$\rho^2 = x^2 + y^2 + z^2$$

Line elements are used for computing path integrals, but that's not how we will use them.

24.3 Volume Element:

In Cartesian: $dV = dx dy dz$

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In Cylindrical: $dV = r dr d\theta dz$

In Spherical: $dV = \rho^2 \sin \phi d\rho d\theta d\phi$

In general: $dV = (h_1 dq_1)(h_2 dq_2)(h_3 dq_3) = \frac{h_1 h_2 h_3 dq_1 dq_2 dq_3}{\frac{\partial(x_1, x_2, x_3)}{\partial(q_1, q_2, q_3)}}$
 $ds = h_1 dq_1 + h_2 dq_2 + h_3 dq_3$

24.4 Example: Cylindrical Coordinates:

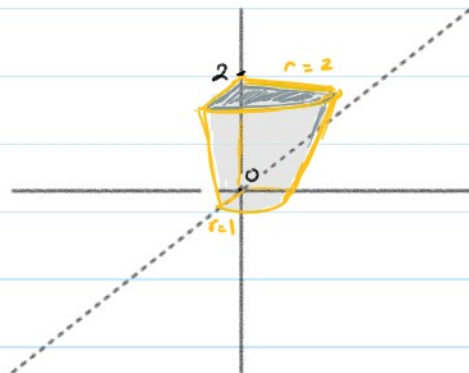
$$ds^2 = dr^2 + r^2 d\theta^2 + dz^2$$

Find the volume of the quarter-frustum

$$1 \leq r \leq 2$$

$$0 \leq \theta \leq \pi/2$$

$$0 \leq z \leq 2$$



Using $V = \iiint dV$

$$= \iiint r dr d\theta dz$$

$$= \int_0^2 \int_0^{\pi/2} \int_1^2 r dr d\theta dz$$

$$= \int_0^2 \int_0^{\pi/2} \left[\frac{r^2}{2} \right]_1^2 d\theta dz$$

$$= \int_0^2 \int_0^{\pi/2} \left(\frac{2^2}{2} - \frac{1^2}{2} \right) d\theta dz$$

$$= (2) \left(\frac{\pi}{2} \right) \left(\frac{3}{2} \right) = \frac{3\pi}{2}$$

24.5 Example:

Suppose, in cylindrical coordinates, we consider a surface with $r = \text{const}$

Then $dr = 0$

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$$\text{so } ds^2 = dr^2 + r^2 d\theta^2 + dz^2 \rightarrow ds^2 = 0^2 + r^2 d\theta^2 + dz^2$$

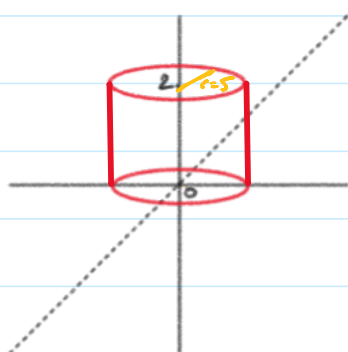
$$\Rightarrow ds^2 = \underbrace{r^2 d\theta^2} + \underbrace{dz^2}$$

Then we get an Area Element: $dA = r d\theta dz$

This can be used to calculate surface areas.

e.g. Find the surface area of the open cylinder.

$$r=5, \quad 0 \leq z \leq 2$$



We have:

$$\begin{aligned} \text{Surface area} &= \iint_R dA = \iint_R r d\theta dz \\ &= \int_0^2 \int_0^{2\pi} 5 d\theta dz = (2)(2\pi)(5) \\ &= 20\pi \end{aligned}$$

Compare with the "high-school" formula: $2\pi r h$

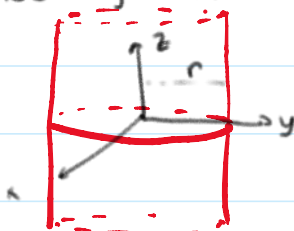
24.6 Special surfaces with one coordinate being constant

In cylindrical coordinates:

$r = \text{const}$

cylinder of radius r

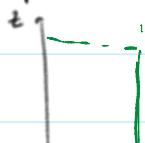
$$dA = r d\theta dz$$

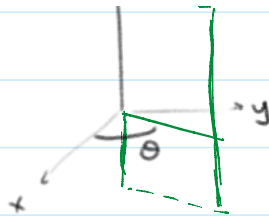


$\theta = \text{const}$

half-plane

$$dA = r dr dz$$

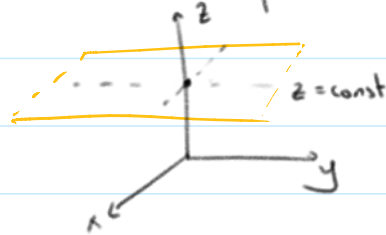




$$z = \text{const}$$

horizontal plane

$$dA = r dr d\theta$$

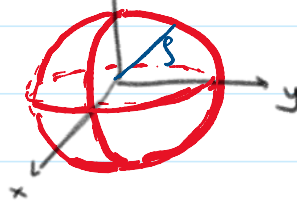


In Spherical Coordinates:

$$\rho = \text{const}$$

sphere of radius ρ

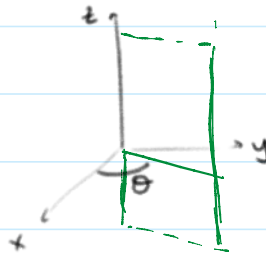
$$dA = \rho^2 \sin \phi d\theta d\phi$$



$$\theta = \text{const}$$

half-plane

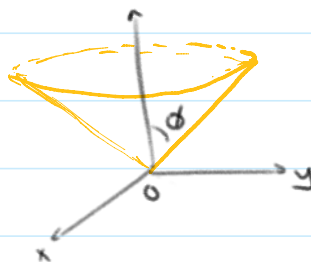
$$dA = \rho d\rho d\phi$$



$$\phi = \text{const}$$

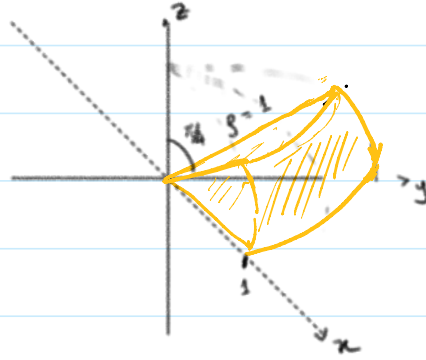
half-cone

$$dA = \rho \sin \phi d\rho d\theta$$



24.7 Example: Find the total surface area of the following wedge of cheese:

Example: Find the total surface area of the following wedge of cheese:



$$0 \leq \rho \leq 1$$

$$0 \leq \theta \leq \pi/2$$

$$\pi/4 \leq \phi \leq \pi/2$$

Split into multiple surfaces:

①

$$\rho = 1$$

$$0 \leq \theta \leq \pi/2$$

$$\pi/4 \leq \phi \leq \pi/2$$



$$so \ dA = \rho^2 \sin \phi \ d\theta \ d\phi$$

$$= 1^2 \sin \phi \ d\theta \ d\phi = \sin \phi \ d\theta \ d\phi$$

$$and \ so \ A = \int_0^{\pi/2} \int_{\pi/4}^{\pi/2} \sin \phi \ d\phi \ d\theta$$

$$= \pi/2 \left[-\cos \phi \right]_{\pi/4}^{\pi/2} = \pi/2 \left(0 + \frac{1}{\sqrt{2}} \right)$$

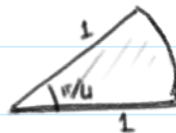
$$= \frac{\pi}{2\sqrt{2}}$$

②

$$\theta = 0 \quad \& \quad \theta = \pi/2$$

$$0 \leq \rho \leq 1$$

$$\pi/4 \leq \phi \leq \pi/2$$



$$dA = \rho \ d\rho \ d\phi$$

$$so \ A = \int_{\pi/4}^{\pi/2} \int_0^1 \rho \ d\rho \ d\phi$$

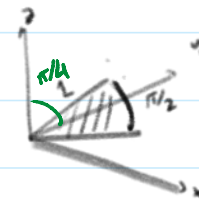
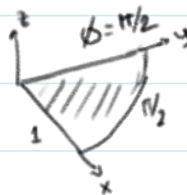
$$= \pi/4 \left[\frac{\rho^2}{2} \right]_0^1 = \frac{\pi}{8} \text{ (same for each } \theta \text{)}$$

③

$$\phi = \pi/4 \quad \& \quad \phi = \pi/2$$

$$0 \leq \rho \leq 1$$

$$0 \leq \theta \leq \pi/2$$



$$dA = \rho \sin \phi \ d\rho \ d\theta$$

$$so \ A = \int_0^{\pi/2} \int_0^1 \rho \sin \phi \ d\rho \ d\theta$$

$$= \left(\frac{\pi}{2} \right) \left(\frac{\rho^2}{2} \right) \Big|_0^1 \sin \phi$$

$$= \pi/4 \sin \phi$$

For $\phi = \pi/2$

$$A = \pi/4 \sin \left(\pi/2 \right) = \pi/4$$

for $\phi = \pi/4$

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$$A = \pi/4 \sin(\pi/4) = \frac{1}{\sqrt{2}} \pi/4$$

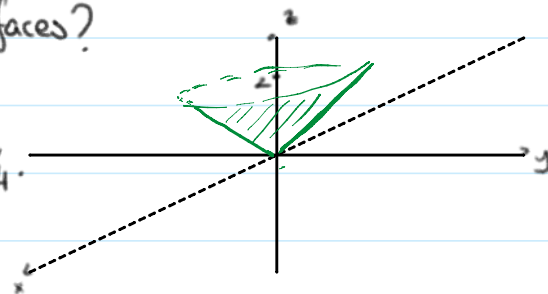
Then total area = $\frac{\pi}{2\sqrt{2}} + 2 \times \frac{\pi}{8} + \frac{\pi}{4} + \frac{\pi}{4\sqrt{2}} = \frac{\pi}{2} + \frac{3\pi}{4\sqrt{2}} = \frac{\pi}{2} \left(1 + \frac{3}{2\sqrt{2}}\right)$

24.8 Does this trick work on other types of surfaces?

Consider the cone: $z = \sqrt{x^2 + y^2}$, $0 \leq z \leq 2$

In spherical coordinates, this is just $\phi = \pi/4$.

In cylindrical coordinates, this is $z = r$.



If we use spherical coordinates, then largest value of

$$\rho^2 = x^2 + y^2 + z^2 = 2^2 + 2^2 = 8 \Rightarrow \rho_{\max} = 2\sqrt{2}$$

So this surfaces is:

$$\left. \begin{array}{l} \phi = \pi/4 \\ 0 \leq \rho \leq 2\sqrt{2} \\ 0 \leq \theta \leq 2\pi \end{array} \right\}$$

for which $dA = \rho \sin \phi d\rho d\theta = \rho \sin(\pi/4) d\rho d\theta$
 $\Rightarrow dA = \frac{1}{\sqrt{2}} \rho d\rho d\theta$

$$\text{Thus } A = \int_0^{2\pi} \int_0^{2\sqrt{2}} \frac{1}{\sqrt{2}} \rho d\rho d\theta = \frac{1}{\sqrt{2}} (2\pi) \left[\frac{\rho^2}{2} \right]_0^{2\sqrt{2}} = \frac{1}{\sqrt{2}} (2\pi) \left(\frac{8}{2} \right)$$

$$\Rightarrow A = \frac{8\pi}{\sqrt{2}} = 4\sqrt{2}\pi \quad \text{Compare with } \pi r l$$

Using Cylindrical coordinates: $ds^2 = dr^2 + r^2 d\theta^2 + dz^2$

The surface is: $r = z$, $0 \leq z \leq 2$, $0 \leq \theta \leq 2\pi$

Then $dr = dz$, which means: $dr^2 = dz^2$

$$ds^2 = dr^2 + r^2 d\theta^2 + dz^2 = dz^2 + z^2 d\theta^2 + dz^2 = 2dz^2 + z^2 d\theta^2$$

$$\Rightarrow ds = \sqrt{2} dz + z d\theta$$

We thus expect: $dA = \sqrt{2} z dz d\theta$

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$$dA = \sqrt{2} z \, dz \, d\theta$$

and hence:

$$A = \int_0^{2\pi} \int_0^2 \sqrt{2} z \, dz \, d\theta$$

$$= (2\pi) \left(\frac{\sqrt{2} z^2}{2} \Big|_0^2 \right) = (2\pi) \left(\frac{2^2}{\sqrt{2}} \right) = 8\pi/\sqrt{2}$$

So this worked! However, it was not obvious that we had to use $dr^2 = dz^2$ in ds^2 instead of $dr = dz$ in ds !